

This assignment is fairly long, but it deals with a number of (related) crucial notions that pervade modern mathematics. It's impossible to progress in advanced mathematics without mastering these ideas. You probably won't be able to get it all done before the next lecture, but keep coming back to it until you have at least tried all the questions. The concepts and methods covered by this assignment are the key to mathematical thinking, and work on this one assignment will yield wide-ranging benefits not just in mathematics but in many others parts of your life.

1. Build a truth table to prove the claim I made earlier that $\phi \Leftrightarrow \psi$ is true if ϕ and ψ are both true or both false, and $\phi \Leftrightarrow \psi$ is false if exactly one of ϕ, ψ is true and the other false. (To constitute a proof, your table should have columns that show how the entries for $\phi \Leftrightarrow \psi$ are derived, one operator at a time.)

2. Build a truth table to show that

$$(\phi \Rightarrow \psi) \Leftrightarrow (\neg\phi \vee \psi)$$

is true for all truth values of ϕ and ψ . A statement whose truth values are all T is called a *logical validity*, or sometimes a *tautology*.

3. Build a truth table to show that

$$(\phi \not\Rightarrow \psi) \Leftrightarrow (\phi \wedge \neg\psi)$$

is a tautology.

4. The ancient Greeks formulated a basic rule of reasoning for proving mathematical statements. Called *modus ponens*, it says that if you know ϕ and you know $\phi \Rightarrow \psi$, then you can conclude ψ .

- (a) Construct a truth table for the logical statement

$$[\phi \wedge (\phi \Rightarrow \psi)] \Rightarrow \psi$$

- (b) Explain how the truth table you obtain demonstrates that *modus ponens* is a valid rule of inference.

5. [This question has a long set-up. The question itself is the very last sentence. TAKE YOUR TIME.]

One way to prove that

$$\neg(\phi \wedge \psi) \quad \text{and} \quad (\neg\phi) \vee (\neg\psi)$$

are equivalent is to show they have the same truth table:

ϕ	ψ	$\phi \wedge \psi$	*	$\neg\phi$	$\neg\psi$	*
			$\neg(\phi \wedge \psi)$			$(\neg\phi) \vee (\neg\psi)$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

Since the two columns marked * are identical, we know that the two expressions are equivalent.

Thus, negation has the affect that it changes $\vee$ into $\wedge$ and changes $\wedge$ into $\vee$. An alternative approach way to prove this is to argue directly with the meaning of the first statement:

1. $\phi \wedge \psi$ means both ϕ and ψ are true.
2. Thus $\neg(\phi \wedge \psi)$ means it is not the case that both ϕ and ψ are true.
3. If they are not both true, then at least one of ϕ, ψ must be false.

4. This is clearly the same as saying that at least one of $\neg\phi$ and $\neg\psi$ is true. (By the definition of negation).
5. By the meaning of *or*, this can be expressed as $(\neg\phi) \vee (\neg\psi)$.

Provide an analogous logical argument to show that $\neg(\phi \vee \psi)$ and $(\neg\phi) \wedge (\neg\psi)$ are equivalent.

6. By a *denial* of a statement ϕ we mean any statement equivalent to $\neg\phi$. Give a useful (and hence natural sounding) denial of each of the following statements.
 - (a) 34,159 is a prime number.
 - (b) Roses are red and violets are blue.
 - (c) If there are no hamburgers, I'll have a hot dog.
 - (d) Fred will go but he will not play.
 - (e) The number x is either negative or greater than 10.
 - (f) We will win the first game or the second.
7. Show that $\phi \Leftrightarrow \psi$ is equivalent to $(\neg\phi) \Leftrightarrow (\neg\psi)$.
8. Construct truth tables to illustrate the following:
 - (a) $\phi \Leftrightarrow \psi$
 - (b) $\phi \Rightarrow (\psi \vee \theta)$
9. Use truth tables to prove that the following are equivalent: $\phi \Rightarrow (\psi \wedge \theta)$ and $(\phi \Rightarrow \psi) \wedge (\phi \Rightarrow \theta)$
10. Verify the equivalence in the previous question by means of a logical argument. (So, you must show that assuming ϕ and deducing $\psi \wedge \theta$ is the same as both deducing ψ from ϕ and θ from ϕ .)
11. Use truth tables to prove the equivalence of $\phi \Rightarrow \psi$ and $(\neg\psi) \Rightarrow (\neg\phi)$.
 $(\neg\psi) \Rightarrow (\neg\phi)$ is called the *contrapositive* of $\phi \Rightarrow \psi$. The logical equivalence of a conditional and its contrapositive means that one way to prove an implication it is to verify the contrapositive. This is a common form of proof in mathematics that we'll encounter later.
12. Write down the contrapositives of the following statements:
 - (a) If two rectangles are congruent, they have the same area.
 - (b) If a triangle with sides a, b, c (c largest) is right-angled, then $a^2 + b^2 = c^2$.
 - (c) If $2^n - 1$ is prime, then n is prime.
 - (d) If the Yuan rises, the Dollar will fall.
13. It is important not to confuse the contrapositive of a conditional $\phi \Rightarrow \psi$ with its *converse* $\psi \Rightarrow \phi$. Use truth tables to show that the contrapositive and the converse of $\phi \Rightarrow \psi$ are not equivalent.
14. Write down the converses of the four statements in question 12.